

E-Commerce Website Using Microservices Architecture

Project Overview

The E-Commerce Website is an online shopping platform designed to provide users with a convenient and efficient shopping experience. The system will allow customers to create accounts, browse products, search and filter items, manage shopping carts, place orders, make payments, and track order status.

The project will be developed using a microservices architecture, where the major functions of the application will be divided into independent services. Each service will manage its own responsibilities and communicate with other services through REST APIs and asynchronous messaging. This architecture will improve scalability, maintainability, fault isolation, and flexibility.

Team Members

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Main Features

- User registration, login, authentication, and role-based authorization.
- Product listing with categories, prices, descriptions, images, and stock information.
- Product search, sorting, and filtering.
- Shopping cart management.
- Wishlist management.
- Order placement and order history.
- Online payment processing or simulated payment functionality.
- Inventory and stock management.
- Product ratings and customer reviews.
- Order status tracking.
- Email or application notifications.
- Administrative dashboard for managing users, products, inventory, and orders.

The system may include separate microservices for users, products, inventory, shopping carts, orders, payments, reviews, and notifications.

Technologies to Be Used

- **Backend:** Java and Spring Boot
- **Microservices:** Spring Cloud, Spring Cloud Gateway, and OpenFeign
- **Security:** Spring Security and JWT
- **Database:** MySQL or PostgreSQL
- **Caching:** Redis
- **Message Broker:** Apache Kafka or RabbitMQ
- **Frontend:** React.js, HTML, CSS, and JavaScript
- **API Documentation:** Swagger/OpenAPI
- **Containerization:** Docker and Docker Compose
- **Version Control:** Git and GitHub